Balasubramaniam Srinivasan

Senior Applied Scientist & Tech Lead, Amazon AGI | Reinforcement Learning · LLM Alignment & Safety · Agentic AI

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Summary

Senior Applied Scientist and Tech Lead at Amazon working on reinforcement learning (RL), post-training, and safety for large language model (LLM) agents. At Amazon AGI (Responsible AI), build and mutate RL environments to study *agentic misalignment*: when LLM agents act as insider threats, reward-hack, or go beyond what was legitimately asked, and how that behavior transfers across workflows. Previously led research-to-production science for agentic AI products shipped from prototype to GA (AgentCore Optimization, AWS Agent Registry, AgentCore Evaluations, Bedrock Intelligent Prompt Routing) used by 1,000+ enterprise customers, including a model router serving 1M+ requests / week at 40%+ lower cost with quality parity. PhD in Computer Science (Purdue); 20+ papers and patents, including at ICLR, NeurIPS, ICML, and EMNLP.

Experience

Amazon · Senior Applied Scientist & Tech Lead (promoted from Applied Scientist, Apr 2024) Jun 2022 – Present

Amazon AGI · AWS Agentic AI (Bedrock AgentCore) · Amazon Bedrock · AWS AI Research & Education. Santa Clara, CA.

Amazon AGI, Responsible AI: RL Environments for Agentic Safety (current)

- *Insider-threat RL environments.* Build RL training environments, graders, and reward signals that measure *agentic misalignment*: LLM agents acting beyond what was legitimately asked, driven by reward hacking and specification gaming.
- *Red-vs-blue environment mutation.* Adversarially mutate existing long-horizon gyms (red vs. blue team) and measure whether mutations raise attack success rate (ASR) at eliciting insider-threat behavior in coding, data, and tool-use workflows.
- *Cross-workflow transfer.* Study how these behaviors, once learned in one workflow, generalize to held-out workflows (misalignment generalization).

AWS Agentic AI, Amazon Bedrock AgentCore: Tech Lead, Agent Optimization, Search & Security

- *AgentCore Optimization (GA Jun 2026, Tech Lead).* Led science for improving production LLM agents from their OpenTelemetry (OTEL) traces: system-prompt and tool-description recommendations validated by batch evaluations and live A/B tests, failure insights, auto-generated skill files, and RL policies for correctness and cost.
- *AWS Agent Registry (GA Aug 2026, Tech Lead).* Led science for discovering and governing *compound agentic systems* (agents, MCP servers, tools, skills) with hybrid search and fine-tuned retrievers. Built a pre-deployment security scanner and *RL-driven automated red-teaming* (prompt injection, jailbreaks, unsafe tool use) mapped to OWASP Agentic Top 10 threat models. **Adopted by 1,000+ enterprise customers.**
- *AgentCore Evaluations (GA Mar 2026, Core Contributor).* Owned synthetic-data generation for agent evaluation and the meta-evaluation of LLM judges (calibration, drift, agreement with human evaluation); contributed task-success, trajectory, tool-use, and safety / policy-adherence methodology. **Adopted by 500+ enterprise customers.**

Amazon Bedrock: Tech Lead, Intelligent Prompt Routing

- *Bedrock Intelligent Prompt Routing (GA Apr 2025, Tech Lead).* Led science for routing prompts across LLMs to maximize quality under user-set cost constraints: calibrated quality predictors trained on 1.5M annotated prompts, a user-controlled quality / cost frontier, and offline / online evaluation (IPR, EMNLP 2025). **1M+ requests / week; 40%+ cost reduction at quality parity with the top model; <150 ms latency.**
- *Bedrock Prompt Optimization (GA Apr 2025, Core Contributor).* Built prompt rewrite strategies and per-task evaluation across Claude, Llama, Nova, and Mistral; co-authored the EMNLP 2025 APO survey.

AWS AI Research & Education, Amazon DataZone: Tech Lead, LLMs for Data Cataloging

- *AutoDoc, Amazon DataZone (GA Mar 2024, Tech Lead).* Led science for LLM-generated data-catalog documentation *without access to column contents*: schema-aware routing, hallucination filtering, LLM-as-a-judge evaluation (US Patent 12,524,628). **100K+ tables across 25+ enterprise domains.**
- *Business Name Generation (GA Sep 2023, Core Contributor).* Designed a small permutation-equivariant model that jointly names all columns of a table; beat state-of-the-art LLMs by 10+ absolute points.

Leadership

- Technical leadership: set technical direction and multi-quarter science roadmaps (presented to Directors / VPs); led cross-functional science and engineering teams from concept to product launch; hiring-committee member for senior-scientist loops; mentored 15+ scientists and PhD interns.

Purdue University · Graduate Research Assistant (PhD) Aug 2018 – May 2022

Advisor: Prof. Bruno Ribeiro. Also Applied Scientist Intern, Amazon AWS (Summers 2020, 2021).

- Equivariant and permutation-invariant architectures with expressiveness theory: Janossy Pooling (ICLR 2019), Relational Pooling (ICML 2019), positional vs. structural representations (ICLR 2020), ESAN (ICLR 2022 Spotlight), hypergraph networks (SDM 2021).

ARM · Software Engineer, Performance Modeling & Analysis Jul 2015 – Jul 2016
Memory-subsystem performance modeling for ARM Cortex processors; built a benchmark suite used by silicon partners.

Education

Purdue University · PhD, Computer Science · GPA 4.00 / 4.00 Aug 2018 – May 2022
Thesis: *On Higher-Order Graph Representation Learning.* Advisor: Prof. Bruno Ribeiro.
University of California, San Diego · M.S., Computer Science · GPA 3.97 / 4.00 Sep 2016 – Jun 2018
BITS Pilani · B.E. (Hons.), Electrical & Electronics Engineering · GPA 8.97 / 10.0 Aug 2011 – May 2015
Top 2% Merit Scholarship; B.E. thesis at the Indian Institute of Science.

Selected Publications and Patents

20+ papers and patents (incl. US Patent 12,524,628), 2,000+ citations. Full list on Google Scholar.

BayesFlow: A Probability Inference Framework for Meta-Agent Assisted Workflow Generation. *EACL 2026 Findings.* B. Yuan, Y. Zhou, Z. Xu, K. Ramnath, A. Feng, **B. Srinivasan**.

MEDAL: Diffusion Language Model Inference with Monte Carlo Tree Search. *EACL 2026 Findings.* Z. Huang, K. Ramnath, Y. Chen, A. Feng, S. Woo, **B. Srinivasan**, Z. Xu, K. Zhou, S. Wang, H. Ding, L. L. Cheong.

IPR: Intelligent Prompt Routing with User-Controlled Quality-Cost Trade-offs. *EMNLP 2025 Industry.* A. Feng, **B. Srinivasan**, *et al.* (Amazon AWS). *Powers Bedrock Intelligent Prompt Routing.*

A Systematic Survey of Automatic Prompt Optimization Techniques. *EMNLP 2025.* K. Ramnath, *et al.* (incl. **B. Srinivasan**), Amazon AWS.

CoverICL: Selective Annotation for In-Context Learning via Active Graph Coverage. *EMNLP 2024.* C. Mavromatis, **B. Srinivasan**, Z. Shen, J. Zhang, H. Rangwala, C. Faloutsos, G. Karypis.

TabSyn: Mixed-Type Tabular Data Synthesis with Score-based Diffusion in Latent Space. *ICLR 2024* ***(Oral)***. H. Zhang, J. Zhang, **B. Srinivasan**, Z. Shen, X. Qin, C. Faloutsos, H. Rangwala, G. Karypis.

OpenTab: Advancing Large Language Models as Open-domain Table Reasoners. *ICLR 2024.* K. Kong, J. Zhang, Z. Shen, **B. Srinivasan**, C. Lei, C. Faloutsos, H. Rangwala, G. Karypis.

HYTREL: Hypergraph-enhanced Tabular Data Representation Learning. *NeurIPS 2023* ***(Spotlight)***. P. Chen, S. Sarkar, L. Lausen, **B. Srinivasan**, S. Zha, R. Huang, G. Karypis.

ESAN: Equivariant Subgraph Aggregation Networks. *ICLR 2022* ***(Spotlight)***. B. Bevilacqua, F. Frasca, D. Lim, **B. Srinivasan**, C. Cai, G. Balamurugan, H. Maron.

Learning over Families of Sets: Hypergraph Representation Learning for Higher Order Tasks. *SDM 2021.* **B. Srinivasan**, D. Zheng, G. Karypis.

On the Equivalence between Positional Node Embeddings and Structural Graph Representations. *ICLR 2020.* **B. Srinivasan**, B. Ribeiro.

Relational Pooling for Graph Representations. *ICML 2019.* R. Murphy, **B. Srinivasan**, V. Rao, B. Ribeiro.

Janossy Pooling: Learning Deep Permutation-Invariant Functions for Variable-Size Inputs. *ICLR 2019.* R. Murphy, **B. Srinivasan**, V. Rao, B. Ribeiro.

Selected Honors and Service

Awards: ICLR 2024 Oral; NeurIPS 2023 and ICLR 2022 Spotlights; UC San Diego Award for Excellence in Teaching (2018).
Area Chair: NeurIPS, COLM, ACL Rolling Review (2023–present). *Reviewer:* ICML, ICLR, AAAI, LoG (2020–present).

Technical Skills

Machine Learning: deep learning, large language models (LLMs), natural language processing (NLP), generative AI, reinforcement learning (RL), Transformers / MoE, diffusion models, graph neural networks, equivariant architectures.

RL & Post-training: PPO, GRPO, DAPO, DPO and preference optimization, RLHF / RLAIF, RLVR, reward modeling, verifiers and rubric-based rewards, fine-tuning (SFT, LoRA / QLoRA / PEFT), distillation, synthetic data generation, test-time search (MCTS).

AI Safety & Alignment: LLM safety and security, RL environments and environment hardening, reward hacking, specification gaming, agentic misalignment, insider-threat and model behavior evaluations, emergent misalignment, model organisms, evaluation awareness, chain-of-thought (CoT) monitoring, automated red teaming, scalable oversight.

Agents & Evaluation: LLM agents, tool use and function calling, MCP, multi-agent (A2A), coding agents, browser and computer-use agents, RAG and hybrid retrieval, LLM-as-a-judge, OpenTelemetry (OTEL), experimental design, ablations, A/B testing, statistical analysis, SWE-bench, tau-bench, WebArena.

Software Development & Systems: Python, C++, SQL; PyTorch, Hugging Face Transformers, TRL, Ray; multi-GPU distributed training (FSDP, DeepSpeed, Megatron-LM, tensor / pipeline parallelism); inference optimization (vLLM, SGLang, TensorRT-LLM, quantization, speculative decoding); large-scale data pipelines and evaluation harnesses; AWS (Bedrock, AgentCore, SageMaker).